

Nikunj Pradhan

nikunj.pradhan31@gmail.com | (314)-719-9396 | St. Peters, MO | nikunjpradhan.com

EDUCATION

Missouri University of Science and Technology

Bachelor of Science, Computer Science

GPA: 4.0/4.0

Aug 2023 - Expected Dec 2026

EXPERIENCE

Full Stack & Data Engineer Co-Op

Jan 2025 – Aug 2025

Hunter Engineering Company

Bridgeton, MO

- Engineered and scaled full-stack applications (FastAPI, React, SQL Server, AWS S3) providing real-time dashboards, data pipelines, and annotation workflows for engineering teams.
- Automated benchmarking workflows by engineering Python CLI tools across AWS and Azure ML pipelines, enabling comparison with vendor models and streamlining team workflow.
- Accelerated distributed multi-GPU training workflows by 66% and development-to-production cycles through close collaboration with R&D teams.
- Created and maintained custom ETL pipelines using FastAPI and Python to clean, validate, and process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL Server and AWS S3.
- Built a CNN-based regression model predicting object coordinates, improving geometric correction accuracy by 23%.

Backend Engineer

June 2025 – July 2025

Tally Receipts

St. Louis, MO

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Integrated FidelSDK for secure card linking in the frontend, designed business logic for safe card tracking in the database, and implemented webhooks to collect user purchase receipts.
- Refactored and modernized Node.js + Express.js services and integrated AWS Lambda, Amplify, and IAM, streamlining backend logic and infrastructure.
- Built AI-powered receipt categorization service, reducing manual entry effort for users and integrating seamlessly with calendar workflows (Google, Microsoft).

Computer Vision Researcher

Aug 2023 – Dec 2024

Missouri University of Science and Technology

Rolla, MO

- Improved CV model accuracy by 62% through optimization, hyperparameter tuning, and algorithmic refinement using C++ OpenCV and Point Cloud.
- Developed a MATLAB/C++ application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling, incorporating advanced CV techniques like watershed segmentation and 3D model triangulation via ray tracing.

PROJECTS

Movie Recommendation & Watchlist Platform | FastAPI, React, PostgreSQL, MinIO, Scikit-learn

- Developed a full-stack movie watchlist platform with secure 2FA login, personalized watchlists, and hybrid recommendations (content-based + ALS collaborative filtering) with per-user caching for accuracy and speed.
- Deployed on Ubuntu using Kubernetes clusters, NGINX reverse proxy, and MinIO for scalable, multi-user media storage and access.

Real-Time File Sharing & Chat Application | React.js, Node.js, Express.js, Socket.IO, MongoDB, GridFS

- Engineered a full-stack real-time chat app with secure JWT authentication, group messaging, and persistent chat history using Socket.IO and MongoDB.
- Implemented scalable file sharing via GridFS and optimized event-based WebSocket communication for sub-500ms message delivery across concurrent sessions.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, MATLAB, Bash, HTML/CSS

Web Development & APIs: FastAPI, React, Node.js, Express.js, WebSockets

Cloud & DevOps: AWS (S3, EC2, Lambda), Azure ML, Docker, Kubernetes, Git, CI/CD, Linux/Ubuntu

Databases: Mongoose, PostgreSQL, Microsoft SQL Server, DynamoDB, Firebase, MinIO

Machine Learning & AI: Pytorch, Scikit-learn, ONNX, CNN/CRNN, Transformers, Deep Learning

Data Science Tools: Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV